last updated : May 29, 2026

Polarization analysis with ^{3}He neutron spin filter on the triple-axis spectrometer PONTA at JRR-3

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Polarized neutron analysis has played an important role in elucidating magnetic order in materials. Conventional measurements have mainly been performed using triple-axis spectrometers with a Cu_2MnAl single-crystal monochromator and a point detector at reactor neutron sources. However, recent interest in complex magnetic orders requires polarization analysis over a wider region of three-dimensional reciprocal space. In this study, we developed a polarized neutron scattering technique that combines a ^{3}He neutron spin filter (^{3}He-NSF) with a two-dimensional detector to analyze scattered neutron polarization over a wide Q region. The method was implemented on the polarized neutron triple-axis spectrometer PONTA at JRR-3 and applied to the multiferroic material Mn_3WO_6. Polarization analysis in three-dimensional reciprocal space extending to high-Q regions successfully resolved the spin-dependent intensities of magnetic reflections distributed outside the scattering plane. These results demonstrate the effectiveness of the present method for analyzing complex magnetic structures.

I. INTRODUCTION

Polarized neutron scattering is a powerful technique for investigating magnetic structures and spin correlations because of the intrinsic spin and magnetic moment of neutrons [1]. In particular, longitudinal polarization analysis, which analyzes the neutron spin states before and after scattering enables the separation of magnetic and nuclear scattering, the separation of coherent and incoherent scattering, and the determination of magnetic moment directions [2]. In reactor-based neutron experiments, such polarization analysis is typically performed using the (111) plane of the ferromagnetic crystal Cu_2MnAl as the neutron polarizer and analyzer at triple-axis spectrometers[3–7], as shown in Fig. 1(a). While this type of instrument has been widely used for magnetic structure analysis, the accessible Q region for analyzing scattered neutrons is limited to the scattering plane. Therefore it has been mainly applied to relatively simple magnetic structures.

In recent years, topologically nontrivial magnetic textures, such as magnetic skyrmions and spin-hedgehog lattices, have attracted considerable attention because their complex spin arrangements can give rise to emergent electromagnetic responses and other unconventional physical phenomena [8, 9]. These magnetic orders are often characterized by a superposition of multiple magnetic modulation wavevectors (multiple q-vectors). As a result, the corresponding magnetic reflections are not confined to a single scattering plane but are distributed over three-dimensional reciprocal space.

Since the discovery of the magnetic skyrmion lattice in MnSi by small-angle neutron scattering (SANS) [8], SANS has played a central role in identifying long-period magnetic structures by detecting their magnetic modulation wavevectors [10–12]. In addition, polarized SANS, which is combined with a large-solid-angle neutron spin analyzer, such as using polarized ^{3}He (^{3}He neutron spin filter, ^{3}He-NSF) has become a powerful technique for analyzing such complex magnetic structures [13, 14]. However, the accessible Q range in SANS experiments is generally limited to relatively low Q. To investigate broader range of complex magnetic structures, it is desirable to extend polarization analysis to higher-Q regions. Thus, we propose a polarization analysis technique in which a ^{3}He-NSF and a two-dimensional detector are placed at arbitrary scattering angles, as shown in Fig. 1(b). Note that this type of instrument has already been reported at several neutron facilities, such as the POLI at MLZ [15], and WOMBAT at ANSTO [16]. However, to the best of our knowledge, quantitative experimental results demonstrating polarization analysis with this type of setup have not yet been reported.

In this work, we implemented polarization analysis using a ^{3}He-NSF combined with a two-dimensional position-sensitive detector on the Polarized neutron triple axis spectrometer, PONTA at JRR-3. Polarized neutron diffraction experiments on the multiferroic material

Mn_3WO_6 was performed using this setup. The performance of the ^{3}He-NSF was evaluated from the obtained nuclear reflection data. Furthermore, three-dimensional reciprocal-space mapping data are presented to show that the present method enables polarization analysis of complex magnetic structures distributed over three-dimensional reciprocal space.

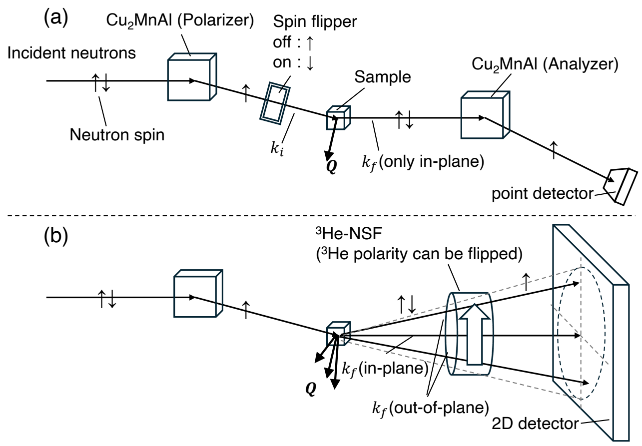

FIG. 1. Schematic illustrations comparing neutron polarization analysis techniques. (a) Conventional setup using a Cu_2MnAl crystal for both polarizer and analyzer, where only in-plane scattered neutrons are detected by a point detector. (b) Polarization analysis setup using a ^{3}He-NSF combined with a two-dimensional detector, enabling detection of both in-plane and out-of-plane scattered neutrons.

II. EXPERIMENTAL DETAILS

A. Setup

PONTA is one of the polarized neutron diffractometer in JRR-3, and it has a longitudinal polarization analysis option [7]. In this option, both incident and scattering neutrons are polarized and analyzed by Cu_2MnAl Heusler crystal monochromator, which provides a neutron wavelength of 2.3 Å. In this experiment, the spin analyzer of the Cu_2MnAl Heusler crystal monochromator and point detector were not used, and ^{3}He-NSF and two-dimensional detector were installed instead. Figure 2 shows a detail of the experimental setup from the sample environment to the detector. The neutron detector was a scintillation-type two-dimensional detector with an active area of 256 mm × 256 mm using wavelength-shifting fiber [17] . The detector was located at an angle of approximately 32° relative to the incident neutron beam direction. A sample to detector distance was approximately 0.9 m. A ^{3}He-NSF was placed as close to the sample as possible to maximize the solid-angle coverage. The angular coverage of the ^{3}He-NSF was 4.7°, allowing polarization analysis over approximately half of the detector area around the detector center. To suppress the background level, a neutron guide tube covered with B_4C shielding tube was installed between the ^{3}He-NSF and the detector. The sample was mounted in a cryostat, and the rotation angles were independently adjustable to align the target reflection with the detector. The Helmholtz coil was used to maintain the neutron spin polarization by applying a vertical magnetic field of approximately 1 mT. The magnetic field in the ^{3}He-NSF was oriented parallel to the beam axis, and spins of the neutrons scattered from the sample adiabatically followed the field direction. A single-crystal sample of Mn_3WO_6 with a volume of 8 mm^3 was mounted on an aluminum holder and installed in a cryostat. The scattering plane was set to the $(H0L)$ plane.

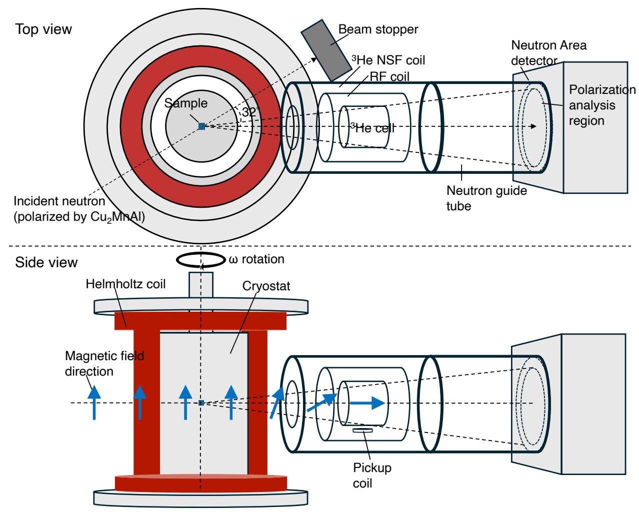

FIG. 2. Schematic illustration of a polarization analysis method enabling broad coverage of reciprocal lattice points in three-dimensional reciprocal space.

B. ^{3}He neutron spin filter

The neutron polarization principle of the ^{3}He-NSF is based on the strong spin dependence of the neutron absorption cross section of ^{3}He nuclei. Neutrons with spins antiparallel to the ^{3}He nuclear spins are absorbed strongly, while those with spins parallel to the ^{3}He nuclear spins are transmitted. The ^{3}He polarization was achieved using Spin-Exchange Optical Pumping (SEOP) method. In this method, circularly polarized laser light polarizes alkali-metal atoms, and the polarization is then transferred to ^{3}He nuclei through spin-exchange collisions. An darkroom environment using the SEOP laser was constructed in the beamline experimental area. The ^{3}He gas was polarized in the darkroom and the ^{3}He cell was transported to the beamline, where neutron measurements were performed during the relaxation of the ^{3}He polarization. It is known as an *ex-situ* SEOP method. The SEOP system employed in this study was designed based on that developed for the polarized neutron spec-

TABLE I. Specifications of ^{3}He cells.

Name	$D \times L$ (mm)	Volume (cm^3)	^{3}He density (amg)	N_2 pressure (Torr)	K/Rb (mol)
Kenji	72×67	140	3.2	97	26
Shimpachi	72×65	146	3.1	100	37

trometer POLANO at J-PARC MLF. The SEOP system and the polarization analysis coil are equipped with a fast adiabatic fast-passage nuclear magnetic resonance (AFP NMR) system to flip the ^{3}He polarization. An NMR pickup coil is also placed at the bottom of the ^{3}He cell to monitor the ^{3}He polarization (Fig. 2). Two ^{3}He glass cells made of aluminosilicate glass (GE180) with nearly identical dimensions were used for the experiment. One cell was installed on the polarized neutron experiment, while the other was being polarized in SEOP system, and the two cells were alternately exchanged during the experiment. The details of the cell specifications are described in Table I. In the table, the D and L represent the diameter and length of the ^{3}He glass cell, respectively.

III. RESULTS

In this section, the results of polarized neutron diffraction experiments on Mn_3WO_6 are presented. First, nuclear Bragg reflections of Mn_3WO_6 were used to evaluate the performance of the ^{3}He-NSFand to determine the ^{3}He polarization and relaxation time required for polarization correction. Then, polarization-analysis results for magnetic reflections of Mn_3WO_6 extracted from three-dimensional reciprocal-space data are described.

A. Performance of ^{3}He neutron spin filter

The ^{3}He polarization was estimated from the nuclear Bragg peak intensity of Mn_3WO_6 single crystal sample. Nuclear peak intensities where the incident neutrons polarized by Cu_2MnAl single crystal monochromator and scattering neutrons analyzed by ^{3}He-NSF were given by the following equations:

$$I^{+\pm}(t) = N_0 \Sigma^{++} T_p T_a(t)(1 \pm P_p P_a(t)) \tag{1}$$

where the supersucripts of $I^{+\pm}$ indicate the incident neutron polarization state and the scattered neutron polarization state, respectively, N_0 is a number of incident neutrons comes from the reactor source, Σ^{++} represents the scattering cross section of the $(10\bar{2})$ nuclear reflection of Mn_3WO_6, T_p is the neutron transmission of the Cu_2MnAl monochromator, $T_a(t)$ is the neutron transmission of the ^{3}He-NSF at time t, P_p is the neutron polarization of the Cu_2MnAl monochromator, and $P_a(t)$ is the neutron polarization of the ^{3}He-NSF. By summing I^{++} and I^{+-} in Eq 1, the following expression is obtained:

$$I^{++}(t) + I^{+-}(t) = 2N_0 \Sigma^{++} T_p T_a(t), \tag{2}$$

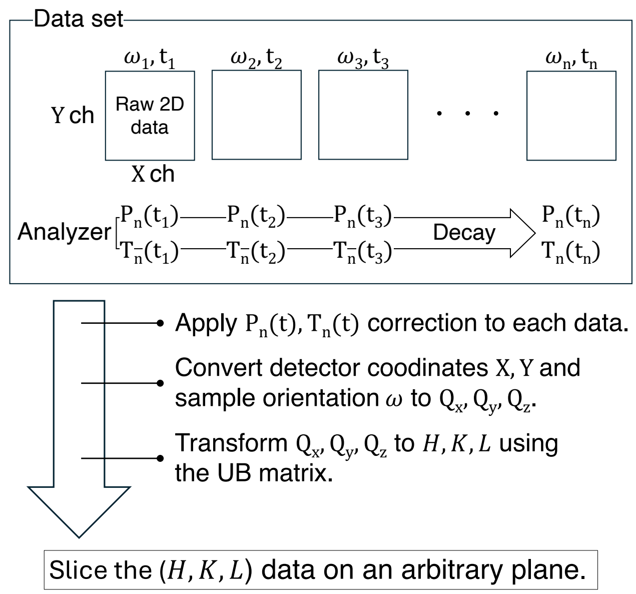

FIG. 3. Schematic diagram of the data-reduction procedure for polarization analysis using an *ex-situ* ^{3}He-NSFand a two-dimensional detector. Raw two-dimensional detector data measured at different sample rotation angles ω_i and times t_i are corrected using the time-dependent neutron polarization $P_n(t_i)$ and transmission $T_n(t_i)$. The detector coordinates (X,Y) and sample rotation angle ω are converted to Q_x, Q_y, Q_z, and then transformed to H, K, L using the UB matrix. The corrected three-dimensional reciprocal-space data can then be sliced on an arbitrary plane.

Here, the measurement time is sufficiently short compared with the relaxation time of the polarized ^{3}He gas, $I^{++}(t)$ and $I^{+-}(t)$ can be regarded as being measured at the same time. The neutron transmission $T_a(t)$ of ^{3}He-NSF, the neutron polarization $P_a(t)$ of ^{3}He-NSF, and ^{3}He polarization $P_{\mathrm{He}}(t)$ are given by the following equations,

$$T_a(t) = T_{\mathrm{cell}} \exp(-\rho d \sigma(\lambda)) \cosh(\rho d \sigma(\lambda) P_{\mathrm{He}}(t)), \tag{3}$$

$$P_a(t) = \tanh(\rho d \sigma(\lambda) P_{\mathrm{He}}(t)), \tag{4}$$

$$P_{\mathrm{He}}(t) = P_0 \exp(-t/\tau), \tag{5}$$

where T_{cell} is the transmission of the ^{3}He glass cell, ρ is a number density of ^{3}He , d is a gas thickness of ^{3}He , $\sigma(\lambda)$ is a neutron absorption cross section of unpolarized ^{3}He , P_0 is the initial polarization at $t = 0$, and τ is the relaxation time. The wavelength λ of the incident neutrons was monochromatized to 2.36 Å by Cu_2MnAl. By taking the ratio with the intensity I_{unpol} measured when the ^{3}He polarization is zero ($P_{\mathrm{He}} = 0$), the following equation is obtained,

$$\frac{I^{++}(t) + I^{+-}(t)}{I_{\mathrm{unpol}}} = 2\cosh(\rho d\sigma(\lambda)P_{\mathrm{He}}(t)). \qquad (6)$$

The ^{3}He gas thickness ρd was determined from the transmission ratio between the ^{3}He cell and an empty cell of the same size and was then used to evaluate ^{3}He polarization $P_{\mathrm{He}}(t)$.

Figure 4 (a,b) shows two-dimensional intensity maps measured just after transporting the ^{3}He cell ($P_{\mathrm{He}}(t = 0)$), where the $(10\bar{2})$ nuclear Bragg peak intensity clearly changes depending on the ^{3}He polarity between up (I^{++}) and down (I^{+-}). Figure 4 (c) shows the result of projecting the peak region of these intensity maps along the x-axis. By integrating the peak intensity and using Eq.6, the ^{3}He polarization was determined to be $P_{\mathrm{He}} = 0.81 \pm 0.05$. Throughout the experiment, nuclear scattering measurements were periodically conducted between magnetic scattering measurements. Figure 5 shows the time evolution of the ^{3}He polarization determined from the nuclear reflection (open circles) and the NMR signal (closed circles). The relaxation time τ was estimated to be $\tau = 130.9 \pm 0.5$ hour. Figure 5 also shows the neutron polarization and transmission calculated using these values. The cell was replaced approximately every two days during the experiment. The validity of the polarization correction was confirmed using the nuclear reflection. The raw intensity profiles of the nuclear reflection show a finite spin-flip intensity owing to the imperfect neutron polarization, as shown in Fig. 6(a). By applying the polarization correction using the obtained ^{3}He polarization, the time-dependent changes in the neutron polarization and transmission of the ^{3}He-NSF were corrected, as shown in Fig. 6(b). These values were used for the polarization correction of the magnetic scattering data. Details of the polarization correction are described in the Appendix A.

B. Application to polarization analysis of Mn_3WO_6

Mn_3WO_6 is a multiferroic material with a ferroaxial crystal structure belonging to the R$\bar{3}$ space group [18]. This system exhibits successive magnetic phase transitions at T = 25 K and 22.5 K, and spontaneous electric polarization has been reported to appear in the low-temperature magnetically ordered phase. In this paper, we focus on the polarization analysis results obtained at 23 K, just below the antiferromagnetic transition. First, using unpolarized neutrons and a two-dimensional detector, intensity maps of antiferromagnetic reflections distributed in the three-dimensional reciprocal space were measured. This measurement was performed using the setup shown in Fig. 2 without the ^{3}He-NSF, and the Cu_2MnAl polarizer was replaced by a PG monochromator. Figure 7(a) shows the reciprocal-space region measured in the present experiment. Figure 7(b) and (d) show intensity maps obtained by integrating the measured intensity along the L direction and the $(-K/2, K, 0)$ direction, respectively. Because the scattering plane was set to the $(H0L)$ plane, a conventional triple-axis setup would mainly provide a reciprocal-space map in this plane, as shown in Fig. 7(d). In the present experiment, the use of a two-dimensional

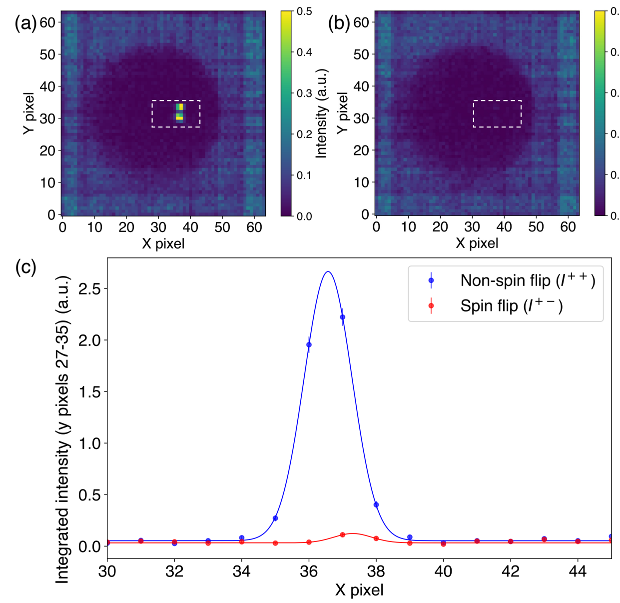

FIG. 4. (a) Non-spin-flip (I^{++}) and (b) spin-flip (I^{+-}) scattering intensity maps. A clear difference in the nuclear Bragg peak intensity is observed upon flipping the ^{3}He polarization. The dashed boxes indicate the integration region. (c) Line profile along the x direction obtained by integrating the intensity within the dashed region in (a) and (b).

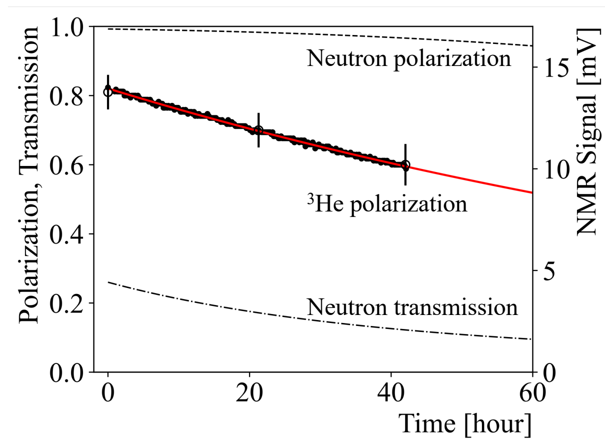

FIG. 5. Time dependence of the ^{3}He polarization, neutron polarization, and neutron transmission. The open circles show the ^{3}He polarization obtained from the intensity of the (102) nuclear Bragg peak, while the closed circles show the NMR signal of ^{3}He . The red line represents the fitting result using an exponential decay function. The dashed line indicates the calculated neutron polarization, and the dot-dashed line indicates the calculated neutron transmission.

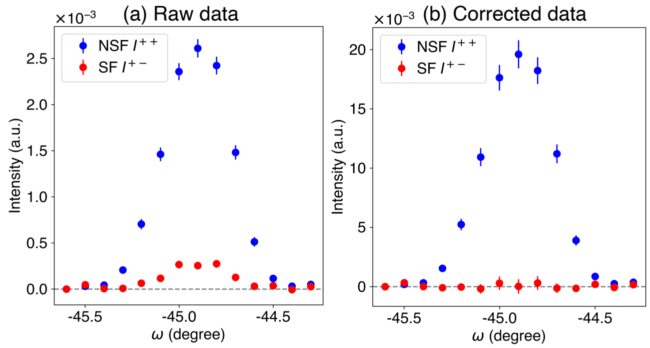

FIG. 6. Intensity of the $(10\bar{2})$ reflection measured while scanning the sample rotation angle ω. The red one shows the I^{++} and blue one shows the I^{+-}. The left shows the raw data, and the right shows the corrected data.

detector enabled the measured three-dimensional data to be projected along different directions. In this projection, magnetic reflections characterized by the three propagation vectors $\boldsymbol{q}_1$, $\boldsymbol{q}_2$, and $\boldsymbol{q}_3$ extending from the $\bar{1}02$ reflection are clearly observed. Polarized neutron scattering measurements were then performed over the same reciprocal-space regions. Figures 7(c) and 7(e) show the polarization-analysis maps obtained from the intensity difference $I^{++} - I^{+-}$. Clear positive and negative contrasts are observed at the magnetic reflection positions identified in the unpolarized intensity maps. In addition, to compare the spin-dependent intensities more quantitatively, line profiles along the $-H$ direction through $\boldsymbol{q}_3$ and $\boldsymbol{q}_1$ are shown in Figs. 7(f) and 7(g), respectively. The relative intensities of the non-spin-flip and spin-flip components are different for the two reflections clearly. Note that the data in Figs. 7(c)-(g) are shown after applying the polarization correction described above. This indicates that the spin-dependent scattering intensities were successfully resolved for magnetic reflections distributed in three-dimensional reciprocal space. The polarization-analysis data provide constraints on the magnetic moment components associated with each propagation vector through the relative non-spin-flip and spin-flip intensities. A detailed magnetic structure analysis based on these constraints will be presented in a separate paper.

IV. SUMMARY AND OUTLOOK

In this work, we developed an experimental approach that combines a ^{3}He-NSF with a two-dimensional detector to perform polarization analysis of arbitrary reciprocal-lattice points distributed in three-dimensional reciprocal space. Although the positions of the ^{3}He-NSF and the two-dimensional detector were fixed in the present experiment, polarization analysis of the incommensurate magnetic reflections of Mn_3WO_6 at 23 K was successfully performed, leading to the determination of its magnetic structure. Recently, a compact *in-situ* ^{3}He-NSF system has been developed at J-PARC, where SEOP is continuously performed during neutron scattering experiments to maintain stable neutron polarization. In future experiments, we plan to install both the compact *in-situ* ^{3}He-NSF system and a two-dimensional detector on a scanning arm. This implementation is expected to provide stable neutron polarization during experiments, extend polarization analysis over a wider region of three-dimensional reciprocal space, and contribute to the elucidation of complex magnetic structures in a broader range of materials.

ACKNOWLEDGMENTS

Appendix A: Polarization correction with *ex-situ* ^{3}He neutron spin filter

In this section, the polarization correction and its error propagation are described based on the polarization matrix method proposed by A. Wildes. The configuration of the neutron scattering apparatus used in this experiment is shown in Fig. 8. The experiment consists of a polarizer and an analyzer. The spin direction of the incident neutrons polarized by the Cu_2MnAl single-crystal monochromator is defined as (+). The neutron spin is preserved by a guide magnetic field. The spin state of the scattered neutrons is then analyzed by a ^{3}He-NSF. The polarization direction of the ^{3}He can be flipped between (+) and (−) using the AFP NMR, and no neutron spin flipper was employed in this experiment. When the ^{3}He-NSF is polarized in the (+) direction, the neutron intensity measured at the detector consists of four contributions, as illustrated in Fig. 8, and can be written as follows.

$$\begin{aligned} I^{++} &= N_0 T_p p_p \Sigma^{++} T_a p_a \\ &+ N_0 T_p p_p \Sigma^{+-} T_a (1 - p_a) \\ &+ N_0 T_p (1 - p_p) \Sigma^{-+} T_a p_a \\ &+ N_0 T_p (1 - p_p) \Sigma^{--} T_a (1 - p_a) \end{aligned} \tag{A1}$$

where N_0 is the number of incident neutrons, T_p,T_a are the total neutron transmission of the polarizer and analyzer, p_p and p_a are the polarization efficiency of the polarizer and analyzer, Σ^{++}, Σ^{+-}, Σ^{-+}, and Σ^{--} are the scattering cross sections for the corresponding spin states. The polarization efficiency p_p and p_a are defined in the range $0.5 \leq p_p,\ p_a \leq 1$, and their relation to the neutron polarization is given by $P_p = 2p_p - 1$ and $P_a = 2p_a - 1$. Only two configurations, I^{++} and I^{+-} can be measured in this experimental setup. Assuming that the numbers of incident neutrons with spin (+) and (−) from the reactor neutron source are equal, Eq. A1 can be simplified into the following matrix form:

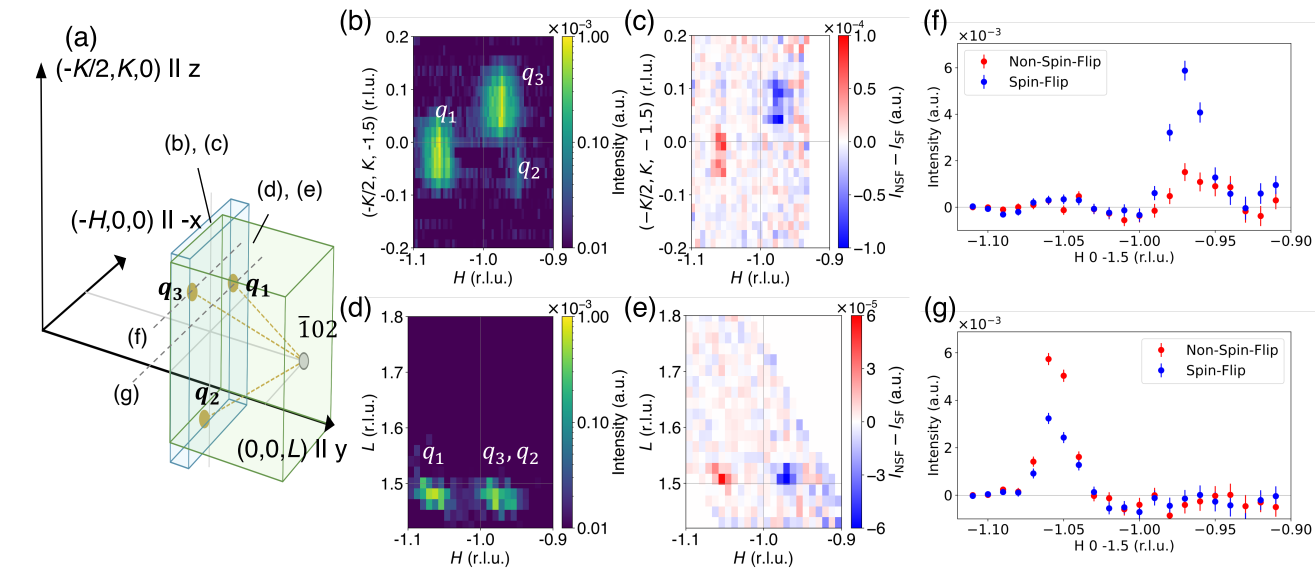

FIG. 7. (a) Schematic illustration of the reciprocal-space region measured in the present experiment. The translucent volumes labeled (b), (c) and (d), (e) indicate the regions integrated along the L direction and the $(-K/2, K, 0)$ direction, respectively. The dashed lines along the $-H$ direction through $\boldsymbol{q}_3$ and $\boldsymbol{q}_1$ indicate the line cuts shown in panels (f) and (g), respectively. (b), (d) Unpolarized neutron intensity maps in the $(H, K/2, -1.5)$ and $(H, 0, L)$ planes, respectively. (c), (e) Corresponding polarization-analysis maps obtained from the difference between the non-spin-flip and spin-flip intensities, $I^{++} - I^{+-}$. (f), (g) Line profiles of the non-spin-flip and spin-flip intensities at $\boldsymbol{q}_3$ and $\boldsymbol{q}_1$, respectively.

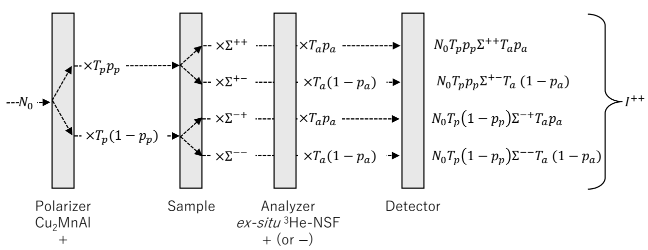

FIG. 8. Schematic diagram of the spin-dependent neutron intensity in the polarized neutron scattering experiment. The incident beam with intensity N_0 is polarized by the Cu_2MnAl polarizer and then scattered by the sample with the spin-dependent cross sections Σ^{++}, Σ^{+-}, Σ^{-+}, and Σ^{--}. The scattered neutrons are analyzed by the *ex-situ* ^{3}He-NSF with transmission T_a and polarization p_a before being detected. The detected intensity I^{++} is given by the sum of the contributions from the four spin-scattering channels when Cu_2MnAl and ^{3}He-NSF are both polarized in the same direction.

$$\begin{bmatrix} I^{++} \\ I^{+-} \end{bmatrix} = \begin{bmatrix} T_p p_p T_a p_a + T_p(1-p_p)T_a(1-p_a) & T_p p_p T_a(1-p_a) + T_p(1-p_p)T_a p_a \\ T_p p_p T_a(1-p_a) + T_p(1-p_p)T_a p_a & T_p p_p T_a p_a + T_p(1-p_p)T_a(1-p_a) \end{bmatrix} \begin{bmatrix} \Sigma^{++} \\ \Sigma^{+-} \end{bmatrix} \tag{A2}$$

Since the *ex-situ* ^{3}He-NSF exhibits a decay of the ^{3}He polarization over time, both the neutron polarization efficiency p_a and the neutron transmission T_a become time dependent. Accordingly, by treating Eq. A2 as a time-dependent matrix equation and taking its inverse, the following expression is obtained.

$$\begin{bmatrix}\Sigma^{++}\\ \Sigma^{+-}\end{bmatrix} = \frac{1}{T_p^2 T_a(t_1)T_a(t_2)(2p_p-1)(p_a(t_1)+p_a(t_2)-1)}\times$$
$$\begin{bmatrix} T_pT_a(t_2)(p_pp_a(t_2)+(1-p_p)(1-p_a(t_2))) & -T_pT_a(t_1)(p_p(1-p_a(t_1))+(1-p_p)p_a(t_1)) \\ -T_pT_a(t_2)(p_p(1-p_a(t_2))+(1-p_p)p_a(t_2)) & T_pT_a(t_1)(p_pp_a(t_1)+(1-p_p)(1-p_a(t_1))) \end{bmatrix}\begin{bmatrix}I^{++}(t_1)\\ I^{+-}(t_2)\end{bmatrix}$$
$$= \frac{1}{T_p^2 T_a(t_1)T_a(t_2)P_p(P_a(t_1)+P_a(t_2))}\begin{bmatrix} T_pT_a(t_2)(1+P_pP_a(t_2)) & -T_pT_a(t_1)(1-P_pP_a(t_1)) \\ -T_pT_a(t_2)(1-P_pP_a(t_2)) & T_pT_a(t_1)(1+P_pP_a(t_1)) \end{bmatrix}\begin{bmatrix}I^{++}(t_1)\\ I^{+-}(t_2)\end{bmatrix} \quad \text{(A3)}$$

The ^{3}He polarization P_{He}, the neutron transmission T, and the neutron polarization P in the ^{3}He-NSF are given by the following expressions. These quantities are experimentally determined and substituted into Eq. A3. Figure 6 shows the $(10\bar{2})$ reflection intensities at I^{++}, I^{+-} and the results after applying the polarization correction using the experimentally obtained T_a and P_a. The constant neutron transmission T_p of the Cu_2MnAl single-crystal monochromator was absorbed into the incident neutron flux. The non-spin-flip scattering intensity increases while the spin-flip scattering intensity decreases after the correction. These results indicate that the time-dependent decrease in neutron transmission and neutron polarization of ^{3}He-NSF are properly corrected.

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